



UNIVERSITY
OF WOLLONGONG
IN DUBAI

Assignment Cover Sheet

Subject Code:	ECTE331	
Submission Type:	Project	
Assignment Title:	Part 1	
Student Name:	Shannen Godinho	Student Number: 8238182

Introduction

This report documents the design, implementation, and testing of a Java-based simulation for a fault-tolerant autonomous drone navigation system. By receiving altitude data from three separate sensors (Sensor A, Sensor B, and Sensor C), the system uses Triple Modular Redundancy (TMR) to reduce the risks of hardware failure. The controller uses a majority vote mechanism to establish the drone's real height. The program generates random probabilities to replicate real-world hardware abnormalities, notably transmission failures and damaged data. Custom exception handling (SensorReadException, SystemReliabilityException) and fallback logic allow the system to smoothly recover from isolated faults or to initiate a crucial SAFE MODE shutdown if hardware integrity fails altogether. All essential events are dynamically captured in a log file.

1. Event Demonstration and Log File Analysis

1. Majority Decision : The system successfully reads valid data from the sensors and establishes a clear majority without anomalies.

```
--- Reading Cycle: 2 ---  
  
Sensor Readings -> A: 119m | B: 119m | C: 121m  
  
[2026-07-03T14:09:36.249906400] MAJORITY DECISION: Agreement between A and B. Selected Altitude: 119m  
[2026-07-03T14:09:36.250906100] OUTLIER DETECTION: Sensor C variance detected.  
  
Reliability Status: System Stable [0/2]
```

Figure 1 : The voting system accurately spotted the agreement and set the drone's altitude to 119 meters.

- ❖ This event was successfully written to the attached log_5181.txt file.

2. Outlier Detection: Two sensors agree on the altitude, while the third sensor gives a different result.

```
--- Reading Cycle: 2 ---  
  
Sensor Readings -> A: 119m | B: 119m | C: 121m  
  
[2026-07-03T14:09:36.249906400] MAJORITY DECISION: Agreement between A and B. Selected Altitude: 119m  
[2026-07-03T14:09:36.250906100] OUTLIER DETECTION: Sensor C variance detected.  
  
Reliability Status: System Stable [0/2]
```

Figure 2 : Sensors A and C both recorded 119m. The method correctly updated the altitude to 119 meters while identifying Sensor B as an outlier

- ❖ This event was successfully written to the attached log_5181.txt file.

3. Sensor Failure: A sensor creates a custom exception by simulating a total hardware/transmission failure.

```
--- Reading Cycle: 2 ---  
[2026-07-03T14:00:01.733103100] SENSOR FAILURE: Sensor C link lost.  
Sensor Readings -> A: 121m | B: 119m | C: INVALID  
[2026-07-03T14:00:01.733103100] RELIABILITY FAILURE: No majority agreement found among different sensor readings.  
[2026-07-03T14:00:01.734590100] FALLBACK DECISION: Maintaining previous valid altitude: 120m  
Reliability Status: Consecutive Failures [2/2]  
[2026-07-03T14:00:01.735098900] CRITICAL ERROR: Maximum reliability failure threshold reached.  
[2026-07-03T14:00:01.736111100] SAFE MODE ACTIVATED: Stopping execution safety routines.  
>> SAFE MODE SHUTDOWN COMPLETE <<
```

Figure 3 :Sensor C issued a SensorReadException. Sensors A and B were still functional and agreed on 120m, the drone continued flying safely despite the loss of Sensor C.

- ❖ This event was successfully written to the attached log_3549.txt file.
4. Corrupted Reading: A sensor sends a distorted integer number that is fully outside of the valid [0:200] meter altitude range.

```
[2026-07-03T14:00:01.696359600] START: Drone navigation controller is now running.  
--- Reading Cycle: 1 ---  
[2026-07-03T14:00:01.720347300] CORRUPTED READING: Sensor C value out of range (-25m)  
Sensor Readings -> A: 121m | B: 120m | C: INVALID  
[2026-07-03T14:00:01.730634100] RELIABILITY FAILURE: No majority agreement found among different sensor readings.  
[2026-07-03T14:00:01.730634100] FALLBACK DECISION: Maintaining previous valid altitude: 120m  
Reliability Status: Consecutive Failures [1/2]
```

Figure 4 : Sensor C returned a measurement of -25m due to a data corruption incident.

- ❖ This event was successfully written to the attached log_3549.txt file.
5. Fallback Decision: The system encounters a situation in which a majority vote is not achievable.

```
--- Reading Cycle: 3 ---  
[2026-07-03T14:09:36.250906100] SENSOR FAILURE: Sensor B link lost.  
Sensor Readings -> A: 120m | B: INVALID | C: 119m  
[2026-07-03T14:09:36.251905700] RELIABILITY FAILURE: No majority agreement found among different sensor readings.  
[2026-07-03T14:09:36.253415700] FALLBACK DECISION: Maintaining previous valid altitude: 119m  
Reliability Status: Consecutive Failures [1/2]
```

Figure 5 : All three sensors gave entirely different values, impossible to get a majority vote. The controller returned to the last known safe height of 119 m after properly avoiding faulty altitude calculation

- ❖ This event was successfully written to the attached log_5181.txt file.

6. Safe Mode Activated : Two consecutive dependability failures are recorded by the system, requiring an emergency shutdown.

```
--- Reading Cycle: 3 ---
[2026-07-03T14:09:36.250906100] SENSOR FAILURE: Sensor B link lost.
Sensor Readings -> A: 120m | B: INVALID | C: 119m
[2026-07-03T14:09:36.251905700] RELIABILITY FAILURE: No majority agreement found among different sensor readings.
[2026-07-03T14:09:36.253415700] FALLBACK DECISION: Maintaining previous valid altitude: 119m
Reliability Status: Consecutive Failures [1/2]

--- Reading Cycle: 4 ---
[2026-07-03T14:09:36.255430] SENSOR FAILURE: Sensor C link lost.
[2026-07-03T14:09:36.255430] SENSOR FAILURE: Sensor B link lost.
[2026-07-03T14:09:36.256429] SENSOR FAILURE: Sensor A link lost.
Sensor Readings -> A: INVALID | B: INVALID | C: INVALID
[2026-07-03T14:09:36.257427800] RELIABILITY FAILURE: Insufficient valid sensors available to vote.
[2026-07-03T14:09:36.258582800] FALLBACK DECISION: Maintaining previous valid altitude: 119m
Reliability Status: Consecutive Failures [2/2]
[2026-07-03T14:09:36.259599500] CRITICAL ERROR: Maximum reliability failure threshold reached.
[2026-07-03T14:09:36.259599500] SAFE MODE ACTIVATED: Stopping execution safety routines.
>> SAFE MODE SHUTDOWN COMPLETE <<
```

Figure 6 : The application successfully threw the `SystemReliabilityException` when it reached the threshold of two consecutive failures.

- ❖ This event was successfully written to the attached log_5181.txt file.

Conclusion

Overall, the developed Java application fulfils every criterion given out in the project brief. It is evident from examining the produced log files that the Triple Modular Redundancy (TMR) architecture effectively eliminates both hardware-level sensor dropouts and random data corruption. When a sensor malfunctions, the program is prevented from abruptly crashing by encasing the main operation in a try-catch blocks. Lastly, the successive failure tracking serves as a dependable backup plan; if the sensor cluster sustains too much damage to recover, the drone seamlessly enters SAFE MODE to safely complete the flight.